

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****CERTIFICATE OF COMPLIANCE****Note:** This table completed by HERSECC Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

A. General Information

01	Project Name:		02	Date Prepared:	
03	Project Location:		04	Building Front Orientation (deg):	
05	CA City:		06	Number of Dwelling Units with Additions:	
07	Zip Code:		08	Fuel Type:	
09	Climate Zone:		10	Total Conditioned Floor Area (ft ²) (Addition):	
11	Building Type:		12	Slab Area (ft ²):	
13	Project Scope:		14	Fenestration Exceptions:	

B. Opaque Surface Building Insulation Details – Framed (Section 150.2(a) and 150.1(c)1)

Notes:

- Where insulation is installed above the roofing membrane, or above the layer used to seal the roof from water penetration, the insulation shall have a maximum water absorption of 0.3 percent by volume when tested according to American Society for Testing and Materials (ASTM) Standard C272.
- Extensions of existing wood-framed walls may retain the dimensions of the existing walls and shall install cavity insulation of R-15 in a 2x4 framing, and R-21 in a 2x6 framing.

01	02	03	04	05	06a	06b	07	08	09	1008	1109
Tag/ID	Assembly Type	Frame Type	Frame Depth (inches)	Frame Spacing (inches)	Proposed			Appendix JA4 Reference		Required	Comments
					Cavity R-value	Continuous Insulation R-value	U-Factor	Table	Cell	U-Factor from Table 150.1-A	

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**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****C.- Opaque Surface Building Insulation Details – Nonframed** (Section 150.1(c)1)

Note:

Where insulation is installed above the roofing membrane, or above the layer used to seal the roof from water penetration, the insulation shall have a maximum water absorption of 0.3 percent by volume when tested according to American Society for Testing and Materials (ASTM) Standard C272.

01	02	03	04	05	06	07	08	09	1008	1109
Tag/ID	Assembly Type	Assembly Materials	Thickness (inches)	Proposed		U-Factor	Joint Appendix JA4 Reference		Required	Comments
				Core Insulation R-value	Continuous Insulation R-value		Table	Cell	U-Factor from Table 150.1-A	

D. Opaque Surface Building Insulation Details – Masonry/Mass Walls (Section 150.1(c)1Bii)

Note: When insulation is added to the outside of a mass wall and/or when the inside is furred and insulated, the performance data may be adjusted using Equation 4-4 in the Reference Appendices, Joint Appendix, JA4.

01	02	03	04	05	06	0705	0806	0907	1008	11	12	1309	1410	1511	1612
Tag/ID	Above or Below Grade?	Masonry/ Mass Type	Masonry/ Mass Thickness (inches)	Proposed				Insulation Layer Joint Appendix JA4 Reference		Required					
				Mass Wall Joint Appendix JA4 Reference		Exterior Insulation		Interior Insulation		Exterior Insulation		Interior Insulation			
				Table	Cell	R-value	U-factor	R-value	U-factor	Table	Cell	R-value	U-factor	R-value	U-factor

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**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****E. Slab On Grade/Concrete Raised Floor Insulation (Table 150.1-A)**

Note: Heated slab floors require mandatory slab insulation (see Table 110.8-A).

01	02	03	04	05	06
Floor Type	Proposed		Required		Comments
	Insulation R-value	Insulation U-factor	Insulation R-value	Insulation U-factor	

F. Radiant Barrier (Section 150.1(c)2)

Notes: A radiant barrier is required (for Climate Zones 2-15)

- Radiant barriers shall meet specific eligibility and installation criteria to receive energy credit for compliance with the Building Energy Efficiency Standards for low-rise residential buildings. Refer to Reference Appendices, Residential Appendix, RA4.2.1
- The emittance of the radiant barrier shall be less than or equal to 0.05 as tested in accordance with American Society for Testing and Materials (ASTM) C1371 or ASTM E408.
- For Prescriptive Compliance the attic shall be ventilated to provide a minimum free ventilation area of not less than 1 square foot (ft²) of vent area for each 300 square feet (ft²) of attic floor area with a minimum of ~~40-30~~ percent to ~~no more than 50 percent~~ upper vents. Refer to Reference Appendices, Residential Appendix, RA4.2.1.
- Ridge vents or gable end vents are recommended to achieve the best performance. The material should be cut to allow for full airflow to the venting. Refer to Reference Appendices, Residential Appendix, RA4.2.1.1.

01	02
Radiant Barrier installed below the roof deck and on all gable end walls	Comment

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****G. Roofing Products (Cool Roof) (Section 150.1(c)11)**Notes:

- Exception 1: Any roof area covered by building integrated photovoltaic (PV) panels and solar thermal panels are not required to comply with exempt from the above Cool Roof requirements.
- Exception 2: Roof constructions with weight of 25 pounds per square foot (lb/ft²) are also not required to comply with exempt the Cool Roof requirements.
- Liquid field applied roof coatings must comply with installation criteria from Section 110.8(i)4.

01	02	03	04	05	06	07	08	09	10	11	12	13
Tag/ID	Exception	Roof Pitch	Method of Compliance	Product Type	CRRP Product ID Number	Proposed				Required		
						Initial Solar Reflectance	Aged Solar Reflectance	Thermal Emittance	SRI (Optional)	Aged Solar Reflectance	Thermal Emittance	SRI (Optional)

H. Fenestration/Glazing Allowed Areas and Efficiencies (Section 150.2(a)1)

01	02	03	04	05	06	07	08	09	10
Addition Type ft²	Maximum Allowed Fenestration Area for All Orientations ft²		Maximum Allowed West-Facing Fenestration Area Only ft²		Maximum Allowed U-factor (Windows)	Maximum Allowed U-factor (Skylights)	Maximum Allowed SHGC (Windows)	Maximum Allowed SHGC (Skylights)	Comments
	The Greater		The Greater						
	Maximum Calculated based on Allowed %	Maximum Calculated Allowed ft²	Maximum Calculated based on Allowed %	Maximum Calculated Allowed ft²					

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**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****I. Fenestration Proposed Areas and Efficiencies** (Section 150.2(a))

Note:

If meeting Exception 1 to 150.1(c)3A, New dwelling units with a conditioned floor area of 500 square feet or less in Climate Zone 5 may comply with a maximum U-factor of 0.30.

If meeting Exception ~~12~~ to 150.1(c)3A, Installing less than or equal to 3 square feet (ft²) glass in door or less than or equal to 3 square feet (ft²) tubular skylight, it is assumed to meet the minimum required U-factor ~~of (0.3027)~~ for climate zones 1 through 5, 11 through 14 and 16, and (0.30) for climate zones 6 through 10 and 15 & SHGC of (0.23) for climate zones 2, 4, 6 through 15 and (0.20) for climate zone 15.

If meeting Exception ~~13~~ to 150.1(c)3A, ~~Installing less than or equal to 3 square feet (ft²) tubular skylight,~~ Installing less than or equal to 16 square feet (ft²) tubular skylight, it is assumed to meet the minimum required U-factor ~~(0.5540)~~ & SHGC (0.30). In Climate Zones 1, 3, 5, and 16 there is no SHGC requirement.

Doors with greater than or equal to 25 percent glazing area are considered glazed doors and are treated as fenestration products.

01	02	03	04	05	06	07	08	09	10	11	12	13	14
Tag/ID	Fenestration Type	Frame Type	Dynamic Glazing	Orientation N, S, W, E	Number of Panes	Proposed Fenestration Area (ft ²) (N, S, E)	Proposed West Facing Fenestration Area (ft ²)	Proposed U-factor	Proposed U-factor Source	Proposed SHGC	Proposed SHGC Source	Exterior Shading Device	Combined SHGC from CF1R-ENV-03
15	Total Proposed Fenestration Area												
16	Maximum Allowed Fenestration Area												
17	Compliance Statement:												
18	Total Proposed West-Facing Fenestration Area												
19	Maximum Allowed West-Facing Fenestration Area												
20	Compliance Statement:												
21	Proposed Fenestration U-factor (Windows)												
22	Required Fenestration U-factor (Windows)												
23	Compliance Statement:												
24	Proposed Fenestration SHGC (Windows)												
25	Required Fenestration SHGC (Windows)												
26	Compliance Statement:												
27	Proposed Fenestration U-factor (Skylights)												
28	Required Fenestration U-factor (Skylights)												

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Registration Date/Time:

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29	Compliance Statement:	
30	Proposed Fenestration SHGC (Skylights)	
31	Required Fenestration SHGC (Skylights)	
32	Compliance Statement:	

J. Opaque Swinging Doors to Exterior (Section 150.1(c)5)

Notes:

- Any door with 25 percent or more glass is counted as a fenestration product in Tables H and I.
- Do not include fire-rated doors between garage or unconditioned space and conditioned space.
- If using weighted average to achieve required maximum U-factor, attach CF1R-ENV-02-E.

01	02	03	04	05	06	07
Tag/ID	Area	Proposed U-factor	Proposed U-factor Source	Required Maximum U-factor	Weighted Average (Yes/No)	Comments

K. Space Conditioning (SC) Systems – Heating/Cooling (Section 150.2(b) or Section 150.1(c)7)

01	02	03
Dwelling Unit Name	Dwelling Unit Total CFA = Sum of Existing + Addition (ft ²)	Comments

L. Water Heating Systems (Section 150.2(a)1D)

List water heaters and boilers for both domestic hot water (DHW) heaters and hydronic space heating.

Options:

- A single heat pump water heater. The storage tank shall not be located outdoors and shall be placed on an incompressible, rigid insulated surface with a minimum thermal resistance of R-10. The water heater shall be installed with a communication interface that meets either the requirements of Section 110.12(a) or has an ANSI/CTA-2045-B communication port; or
- A single heat pump water heater that meets the requirements of NEEA Advanced Water Heater Specification Tier 3 or higher; or
- ~~A gas or propane instantaneous water heater with an input of 200,000 Btu per hour or less and no storage tank; or~~

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4.3. For addition that are 500 square feet or less, an ~~instantaneous~~ electric water heater with point of use distribution as specified in the Reference Appendices, Residential Appendix, RA4.4.5

01	02	0302	0403	0504	0605	07	0806	0907
Water Heating System ID or Name	Water Heating System Type	System Option (from §150.2(a)1D)	# of Water Heaters/ Compressors in System	Water Heater Type	Fuel Type	Rated Input	Tank Location	Distribution Type

M. Indoor Air Quality (IAQ) Fan Information

01	02	03
Fan Name	IAQ Type	Comments

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Registration Date/Time:

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**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS**

Documentation Author's Declaration Statement	
1. I certify that this Certificate of Compliance documentation is accurate and complete.	
Documentation Author Name:	Documentation Author Signature:
Company:	Signature Date:
Address:	CEA/ <u>HERSECC</u> Certification Identification (if applicable):
City/State/Zip:	Phone:
Responsible Person's Declaration Statement	
I certify the following under penalty of perjury, under the laws of the State of California: 1. The information provided on this Certificate of Compliance is true and correct. 2. I am eligible under Division 3 of the Business and Professions Code to accept responsibility for the building design or system design identified on this Certificate of Compliance (responsible designer). 3. The energy features and performance specifications, materials, components, and manufactured devices for the building design or system design identified on this Certificate of Compliance conform to the requirements of Title 24, Part 1 and Part 6 of the California Code of Regulations. 4. The building design features or system design features identified on this Certificate of Compliance are consistent with the information provided on other applicable compliance documents, worksheets, calculations, plans and specifications submitted to the enforcement agency for approval with this building permit application. 5. I understand that a registered copy of this Certificate of Compliance shall be made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to accomplish this requirement. 6. I understand that a registered copy of this Certificate of Compliance is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to accomplish these requirements.	
Responsible Designer Name:	Responsible Designer Signature:
Company:	Date Signed:
Address:	License:
City/State/Zip:	Phone:

For assistance or questions regarding the Energy Standards, contact the Energy Hotline at: 1-800-772-3300

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CF1R-ADD-01-E User Instructions

Minimum requirements for prescriptive addition compliance can be found in Building Energy Efficiency Standards Section 150.2(a), and Table 150.1-A. Completing these compliance documents will require that you have the Reference Appendices for the 2022-2025 Building Energy Efficiency Standards, which contain the Joint Appendices used to determine climate zone and to complete the section for opaque surfaces. When the term CF1R is used it means the CF1R-ADD-01. Worksheets are identified by their entire name and subsequently by only the worksheet number, such as CF1R-ENV-02.

Instructions for sections with column numbers and row letters are given separately.

If any part of the addition does not comply, prescriptive compliance fails, in which case the performance (or computer) compliance approach may be used in an attempt to achieve compliance. Only the new construction is required to meet the requirements specified in this documentation. If any alterations to the existing building are occurring, those are documented on one or more of the CF1R-ALT compliance documents.

A. General Information

1. Project Name: Identifying information, such as owner's name.
2. Date Prepared: Date of document preparation.
3. Project Location: Legal street address of property or other applicable identifying information.
4. Building Front Orientation: Building front expressed in degrees, where North = 0, East = 90, South = 180, and West = 270. The Standards (Section 100.1) include the following additional details for determining orientation:
 - North is oriented to within 45 degrees of true north, including 45 degrees east of north;
 - East is oriented to within 45 degrees of true east, including 45 degrees south of east;
 - South is oriented to within 45 degrees of true south, including 45 degrees west of south;
 - West is oriented to within 45 degrees of true west, including 45 degrees north of west.
5. CA City: Legal city/town of property.
6. Number of Dwelling Units with Additions: this field will automatically default to 1 for single family
7. Zip Code: 5-digit zip code for the project location (used to determine climate zone).
8. Fuel Type: ~~Natural Gas, Liquefied Propane Gas, or~~ this field will automatically default to Electricity.

~~NOTE: Prescriptive compliance only allows electricity if natural gas is not connected to the building.~~

9. Climate Zone: From Joint Appendix JA2.1.1.
10. Total Conditioned Floor Area: Enter the new conditioned floor area, in ft², as measured from the outside of exterior walls of the addition.
11. Building Type: this field will automatically default to Single Family (includes duplex)
12. Slab Area: Area of the first floor slab of the addition (if any) in square feet (ft²).

13. Project Scope: Select all that apply: Addition less than 300 square feet (ft²), Addition greater than 300 to less than 400 square feet (ft²), Addition greater than 400 to less than 700 square feet (ft²), Addition greater than 700 to less than 1000 square feet (ft²), Accessory Dwelling Unit (ADU) Addition 300 square feet (ft²) or less~~less than 300 square feet (ft²)~~, Accessory Dwelling Unit (ADU) Addition greater than 300 ft² up to 400 ft²~~greater than 300 to less than 400 square feet (ft²)~~, Accessory Dwelling Unit (ADU) Addition greater than 400 ft² up to 700 ft²~~greater than 400 to less than 700 square feet (ft²)~~, Accessory Dwelling Unit (ADU) Addition greater than 700 ft² up to 1000 ft²~~greater than 700 to less than 1000 square feet (ft²)~~, space heating system, space cooling system, space conditioning duct system, water heating, or fenestration, kitchen remodel.
14. Exceptions to Fenestration U-factor and SHGC: Installing less than or equal to 3 square feet (ft²) glass in door, Installing less than or equal to 3 square feet (ft²) tubular skylight, Installing less than or equal to 16 square feet (ft²) skylight, in Climate Zones 2, 4, and 6 through 15 with a maximum U-factor of 0.40 and a maximum SHGC of 0.30 or NA.

B. Opaque Surface Details - Framed

1. Tag/ID: A label (if any) from the plans, such as A1.4 or wall.
2. Assembly Type: Roof, Ceiling, Wall, or Floor.
3. Frame Type: Wood or Metal.
4. Frame Depth: Nominal dimensions (in inches) of framing material such as 2x4 or 2x6.
5. Frame Spacing: 16, or 24 or 48 (inches on center).
- 6a. Proposed Cavity R-value: Insulation installed between framing members.
- 6b. Proposed Continuous Insulation: R-value of rigid or continuous insulation (not interrupted by framing). See Table 4.3.4. of the Reference Appendices, Joint Appendix, JA4 for metal frame construction.
7. Proposed U-factor: The U-factor for the proposed assembly must be less than or equal to Column 10 or have an attached Area Weighted Average Calculation Worksheet (CF1R-ENV-01-E) to show that a weighted U-factor for multiple assemblies will meet the maximum value in Column 10.
- ~~8. Joint Appendix JA4 Table: Table number used to determine the R-value or U-factor (e.g., an attic assembly is 4.2.1).~~
- ~~9. Joint Appendix JA4 Cell: Cell number used to determine the R-value or U-factor (e.g., an R-38 ceiling with 24-inch on center framing is A21).~~
- 10.8. Required U-factor: From Package A or from Section 150.2. Value required based on climate zone and assembly type.
- 11.9. Comments: Any notes regarding location, unique conditions, or attachments.

C. Opaque Surface Details – Non-Framed

1. Tag/ID: A label (if any) from the plans, for example, A1.4 or wall.
2. Assembly Type: Roof, Wall.
3. Assembly Materials: SIP OSB, SIP I-Joist, SIP Single 2x, SIP Double 2x.
4. Thickness: Thickness in inches.

5. Proposed Core Insulation R-value: Insulation installed within the materials or on the inside. See Reference Appendices, Joint Appendix, JA4 for guidance.
6. Proposed Continuous Insulation R-value: Insulation installed on the exterior. See Reference Appendices, Joint Appendix, JA4 for guidance.
7. Proposed U-factor: Proposed assembly U-factor from Reference Appendices, Joint Appendix, JA4 or CF1R-ENV-02-E. Must be less than or equal to Column 10.
- ~~8. Appendix JA4 Table: Table number used to determine the R value or U factor (e.g., a SIP wall is 4.3.2).~~
- ~~9. Appendix JA4 Cell: Cell number used to determine the R value or U factor (e.g., a 4.5-inch thick Oriented Strand Board (OSB) wall with R-18 core insulation and no continuous insulation is A5).~~
- ~~10-8.~~ Required U-factor from Table 150.1-A: Based on assembly type and climate zone.
- ~~11-9.~~ Comments: Any notes regarding location, unique conditions, or attachments.

D. Opaque Surface Details – Masonry/Mass Walls

1. Tag/ID: A label (if any) from the plans, for example, A1.4 or wall.
2. Mass Wall Location~~Walls Above Grade~~: Indicate whether the mass wall is installed above grade or below grade.~~Yes or No.~~
3. Masonry/Mass ~~Masonry/Wall~~ Type: Clay Brick, Clay Hollow Unit, Concrete Masonry Unit (CMU) Light Weight, Concrete Masonry Unit (CMU) Medium Weight, Concrete Masonry Unit (CMU) Normal Weight, Concrete, ICF. See Reference Appendices, Joint Appendix, JA4 for guidance.
4. Masonry/Mass ~~Masonry/Wall~~ Thickness: Thickness (in inches) of mass.
- ~~5. Joint Appendix JA4 Reference Table: Table number used to determine the R value or U factor (e.g., an Insulating Concrete Form (ICF) wall is 4.3.13).~~
- ~~6. Joint Appendix JA4 Reference Cell: Cell number used to determine the R value or U factor (e.g., an 8-inch thick Insulating Concrete Form (ICF) wall with 2 inches of Expanded Polystyrene (EPS) (R-15.4) is C1).~~
- ~~75-86.~~ Proposed Exterior Insulation R-value or U-factor: Enter the R-value or U-factor of proposed insulation on the outside surface of the mass wall. See Reference Appendices, Joint Appendix, JA4 for guidance. Use the same descriptor (R-value or U-factor) throughout Table D.
- ~~97-108.~~ Proposed Interior Insulation R-value or U-factor: Enter the R-value or U-factor of proposed insulation on the inside surface of the mass wall. See Reference Appendices, Joint Appendix, JA4 for guidance. Use the same descriptor (R-value or U-factor) throughout Table D.
- ~~11. Joint Appendix JA4 Table: Table number used to determine the R value or U factor (e.g., an Insulating Concrete Form (ICF) wall is 4.3.13).~~
- ~~Joint Appendix JA4 Cell: Cell number used to determine the R value or U factor (e.g., an 8-inch thick Insulating Concrete Form (ICF) wall with 2 inches of Expanded Polystyrene (EPS) (R-15.4) is A6).~~~~139-1410.~~ Required Exterior Insulation R-value or U-factor: The required R-value or U-factor (whichever descriptor was selected in Column 7 or 8) for exterior insulation will be completed based on the Table 150.1-A requirements for the wall type.
- ~~1511-1612.~~ Required Interior Insulation R-value or U-factor: The required R-value or U-factor (whichever descriptor was selected in Column 9 or 10) for interior insulation will be completed based on the Table 150.1-A requirements for the wall type.

E. Slab On Grade/Concrete Raised Floor Insulation

Slab edge performance specifications and installation criteria are found in Sections 150.0(l) and 150.1(c)1D (Table 150.1-A). Requirements vary by climate zone and slab conditions.

1. Floor type: Types include slab-on-grade or raised slab.
 - Slab-on-grade floors require slab edge insulation in climate zone 16 only.
 - Raised slab must be insulated to R-8 in climate zones 1, 2, 11, 13, 14 and 16, R-4 in climate zones 12 and 15, and no insulation is required in climate zones 3-10.
2. Proposed R-value: When required, insulation can be specified by either R-value or U-factor (use the same descriptor throughout Table E). When specifying an R-value complete Column 2. Use the same descriptor (R-value or U-factor) throughout Table E.
3. Proposed Insulation U-Factor: When required, specify the U-factor of proposed insulation in Column 3.
4. Required Insulation R-value: Specify the value required, which will vary by climate zone and type of slab. Values are from Table 150.1-A.
5. Required Insulation U-factor: Specify the value required, which will vary by climate zone and type of slab. Values are from Table 150.1-A.
6. Comments: Any notes regarding location, unique conditions, or attachments.

NOTE: There is a mandatory slab edge insulation requirement for heated slab floors. Since mandatory requirements are not listed on the Certificate of Compliance, this is provided for information purposes only. The specific requirements are in Sections 110.8(g) and Table 110.8-A.

F. Radiant Barrier

1. Radiant Barrier installed below the roof deck and on all gable end walls: Yes or No. ~~Radiant barriers are required in climate zones 2-15.~~
2. Comments: Any notes regarding location, unique conditions, or attachments.

NOTE: Radiant barrier performance specifications and installation criteria are found in Sections 110.8(j) and 150.1(c)2, and in Reference Appendices, Residential Appendix, RA4.2.1.

G. Roofing Products (Cool Roof)

Roofing requirements are found in Section 110.8(i) and 150.1(c)11. Depending on the climate zone and roof slope, a cool roof (defined as a minimum aged solar reflectance and thermal emittance, or a minimum SRI) may be required by Package A.

NOTE: Exceptions include (1) additions of 300 square feet (ft²) or less, (2) low-slope roofs (pitch less than 2:12) in climate zones 1-12, 14 and 16; (3) steep-slope roofs (pitch greater than or equal to 2:12) in climate zones 1-9 and 16; (4) roof constructions that have thermal mass over the roof membrane with at least 25 pounds per square foot (lb/ft²); and (5) any roof area covered by building integrated photovoltaic (PV) panels and solar thermal panels (the area of roof not covered by photovoltaic panels would still need to meet any applicable cool roof requirements).

1. Tag/ID: A label (if any) from the plans, such as R1.
2. Exception: User selects from 1, 2, or None. Mass roofs are not required to have a cool roof even if the climate zone specifies minimum performance requirements.

3. Roof Pitch: Expressed as 4:12, for example, which means the roof rises 4 feet within a span of 12 feet. When roofs have multiple pitches the requirements are based on the pitch of 50 percent or more of the roof.
4. Method of Compliance: Indicate if the method of compliance is going to be based on Aged Solar Reflectance and Thermal Emittance or is it going to be based on the Solar Reflectance Index (SRI).
5. Product Type: See Cool Roof Rating Council's directory. Generally, product types include single-ply roof, wood shingles, asphalt roof, metal roof, tile roof.
6. The CRRC Product ID Number is obtained from the [Cool Roof Rating Council's Rated Product Directory](#). Products are listed by manufacturer, brand, type of installation, roofing material, and color, as well as product performance.
7. Proposed Initial Solar Reflectance: Based on the product chosen from the [Cool Roof Rating Council's Rated Product Directory](#). If using default assumption indicate NA since the Aged Solar Reflectance is available.
8. Proposed Aged Solar Reflectance: Value is from the [Cool Roof Rating Council's Rated Product Directory](#). If the aged value is not available, calculate the calculated Aged Solar Reflectance using the Solar Reflectance Index (SRI) Calculation worksheet located on the [California Energy Commission website](#) or the aging equation $\rho_{aged} = [0.2 + \beta(\rho_{initial} - 0.2)]$, where $\rho_{initial}$ = the initial solar reflectance and soiling resistance β is listed by product type below.

VALUES OF SOILING RESISTANCE β BY PRODUCT TYPE

Product Type	CRRC Product Category	β
Field-Applied Coating	Field-Applied Coating	0.65
Other	Not A Field-Applied Coating	0.70

9. Proposed Thermal Emittance: From the product specification default value. If using a calculated SRI, enter the thermal emittance used to calculate SRI.
10. Proposed Solar Reflectance Index (SRI): It is optional to meet the SRI, but if chosen to do so use the Solar Reflectance Index (SRI) Calculation Worksheet found on the [California Energy Commission website](#). ~~Error! Hyperlink reference not valid.~~
11. Required Aged Solar Reflectance: Based on climate zone and roof slope.
12. Required Thermal Emittance: Based on climate zone and roof slope.
13. Required Solar Reflectance Index (SRI): Based on climate zone and roof slope.

If the cool roofing requirements will be met by a liquid field applied coating, Section 110.8(i)4 requires the coating be applied across the entire roof surface and meet the dry mil thickness or coverage recommended by the manufacturer.

H. Fenestration/Glazing Allowed Areas and Efficiencies

Fenestration areas are expressed in square feet, not square inches.

The climate zone and size of the addition will affect the area of fenestration (also known as glazing) allowed. If limited to 20 percent, for example, this is calculated as Conditioned Floor Area (CFA) of the addition x 0.20 = Total square footage (ft²) of fenestration allowed.

1. Addition Type: Based on “Project Scope.” The addition’s area in square feet (ft²)—whether less than or equal to 300, greater than 300 to less than or equal to 400, greater than 400 to less than or equal to 700, or greater than 700 to less than or equal to 1,000.

2-9. These fields will be completed based on conditioned floor area of the addition and/or climate zone. The values in these fields will be entered into Section I.

Maximum allowed fenestration area for all orientations is the greater of the values in Column 2 or 3:

2. Maximum Calculated based on Allowed Percentage (%): The addition’s CFA multiplied by the allowed percentage (%). The maximum total fenestration area is 30 percent for additions up to 400 square feet (ft²), 25 percent for additions greater than 400 square feet (ft²) but no greater than 700 square feet (ft²), and 20 percent for additions greater than 700 square feet (ft²).
3. Maximum Calculated Allowed Area (ft²): The maximum total fenestration area is 75 square feet (ft²) for additions up to 400 square feet (ft²), 120 square feet (ft²) for additions greater than 400 square feet (ft²) but no greater than 700 square feet (ft²), and 175 square feet (ft²) for additions of greater than 700 square feet (ft²).

Maximum allowed west-facing area is the greater of the values in Column 4 or 5:

4. Maximum Calculated based on Allowed Percentage (%): The maximum west-facing fenestration area (in climate zones 2, 4, and 6-16) is 5 percent for additions greater than 700 square feet (ft²).
5. Maximum Calculated Allowed Area (ft²): The maximum west-facing fenestration area (in climate zones 2, 4, and 6-16) is 60 square feet (ft²) for additions no greater than 700 square feet (ft²), and 70 square feet (ft²) for additions of greater than 700 square feet (ft²).

Addition CFA:	≤ 400 ft ²		> 400 to ≤ 700 ft ²		> 700 to ≤ 1,000 ft ²	
	The Greater Of:		The Greater Of:		The Greater Of:	
Orientation	Percentage	Area (ft ²)	Percentage	Area (ft ²)	Percentage	Area (ft ²)
West-facing (CZs 2, 4, 6-16)	-	60	-	60	5%	70
All Orientations	30%	75	25%	120	20%	175

NOTE: West includes any vertical fenestration oriented to within 45 degrees of true west (in either direction), including 45 degrees north of west, any skylights oriented west, and skylights facing any direction with a pitch of less than 1:12.

6. Maximum Allowed U-factor (Windows): Maximum area-weighted average of 0.30 for climate zones 6-10 and 15 all climate zones, and 0.27 for climate zones 1-5, and 11-14 and 16.

7. Maximum Allowed U-factor (Skylights): Maximum area-weighted average of 0.30 for climate zones 6-10 and 15, and 0.27 for climate zones 1-5, 11-14 and 16~~all climate zones~~, unless meeting one of the Exceptions to 150.1(c)3A. If meeting one of the Exceptions, this field will be 0.55~~0.40~~.
8. Maximum Allowed SHGC (Windows): Maximum area-weighted average of 0.23 for climate zones 2, 4, and 6-~~16~~14 and 0.20 for climate zone 15; otherwise N/A.
9. Maximum Allowed SHGC (Skylights): Maximum area-weighted average of 0.23 for climate zones 2, 4, 6 through 15~~and 0.20 for climate zone 15 for all climate zones~~, unless meeting one of the Exceptions to 150.1(c)3A. If meeting one of the Exceptions, this field will be 0.30; otherwise, N/A.
10. Comments: Any notes regarding location, unique conditions, or attachments.

I. Fenestration/Glazing Proposed Areas and Efficiencies

1. Tag/ID: Provide a name or designator for each unique type of fenestration surface. This designator should be used consistently throughout the plan set (elevations, finish schedules, etc.) such as Window-1, Skylight-1, etc. to identify each surface. It should also be consistently used on the other compliance documents.
2. Fenestration Type: Indicate the type of fenestration construction (e.g., Fixed Window, Operable Window, Skylight, Tubular Skylight, or Glass in Door).
NOTE: Doors with glazing are counted in one of two ways. The entire area of a door with 25 percent or more glazing is considered fenestration. A door with less than 25 percent glazing can be considered as all fenestration, or can be calculated as the actual glass area with a 2-inch (0.17 ft) frame all around.
3. Frame Type: Metal, metal thermal break, or non-metal.
4. Dynamic Glazing: Indicate whether the fenestration has an integrated shading device, chromogenic glazing, or none for no dynamic glazing. Chromogenic glazing shall be considered separately from other fenestration types.
5. Orientation: Orientation can be North, East, South, or West. If documentation is for a building that may be built in any direction, in a climate zone that limits west-facing fenestration, complete this section assuming the side of the building with the most fenestration faces west.
NOTE: West includes any vertical fenestration oriented to within 45 degrees of true west, excluding 45 degrees south of west; any skylights oriented west; and skylights facing any direction with a pitch of less than 1:12.
6. Number of Panes: Indicate the number of panes for each Tag/ID; is it a single, double, or triple pane window?
7. Proposed Fenestration Area (ft²): The size of any windows, doors with glass, or skylights within the floor area of the addition (combine windows with the same characteristics). Indicate the area in square feet (ft²) of each exterior fenestration type, including west-facing fenestration.
8. Proposed West Facing Fenestration Area (ft²): In climate zones 2, 4, and 6-16, enter the size of any west-facing windows, doors with glass, or skylights within the floor area of the addition. Indicate the area in square feet (ft²) of each exterior west-facing fenestration type separately.
9. Proposed U-factor: Enter

- (a) the National Fenestration Rating Council (NFRC) U-factor based on the proposed brand and type of fenestration using [NFRC certified values](#); or
- (b) the default value from Table 110.6-A; or
- (c) the NA6.2 alternate default U-factor (for non-rated site-built fenestration only); or
- (d) the area-weighted average from CF1R-ENV-02.

If any products (other than the exceptions noted below) have a higher U-factor than 0.30, first complete a CF1R-ENV-02-E to calculate the area-weighted average U-factor, which must be 0.30 or less, and attach it to the CF1R-ADD-01-E.

NOTES: (1) If meeting Exception 1 to 150.1(c)3A, New dwelling units with a conditioned floor area of 500 square feet or less in Climate Zone 5 may comply with a maximum U-factor of 0.30.

(2) For the exception – up to 3 square feet (ft²) of glass in door, enter 0.30 for climate zones 6 through 10 and 15, and enter 0.27 for climate zones 1 through 5, 11 through 14 and 16. For the exceptions – up to 3 square feet (ft²) of tubular skylights and up to 16 square feet (ft²) of skylight area, enter 0.55.

(23) For the exceptions - up to 3 square feet (ft²) of tubular skylights and up to 16 square feet (ft²) of skylight area, enter 0.40. For the exception – up to 3 square feet (ft²) of glass in door, enter 0.30.

(34) Dynamic glazing is a glazing system that changes its performance U-factor and SHGC based on the physical environment. Dynamic glazing includes chromogenic glazing or integrated shading systems (this does not include internally or externally mounted shading devices). If using dynamic glazing, use the lowest tested U-factor and solar heat gain coefficient (SHGC) in Columns 9 and 11.

10. Source: The source of the U-factor data for the fenestration product—indicate whether National Fenestration Rating Council (NFRC), Tables 110.6-A and 110.6-B, Equations NA6-1 and NA6-2, or Area-weighted Average Worksheet (CF1R-ENV-02).

11. Proposed SHGC: In climate zones 2, 4, and 6-16, enter

- (a) the National Fenestration Rating Council (NFRC) solar heat gain coefficient (SHGC) based on the proposed brand and type of fenestration using [NFRC certified values](#), or
- (b) the default value from Table 110.6-B, or
- (c) the NA6.3 alternate default SHGC (for non-rated site-built fenestration only), or
- (d) the Area-weighted Average from CF1R-ENV-02.

If any products (other than the exceptions noted below) have a higher solar heat gain coefficient (SHGC) than 0.23 in a climate zone with a maximum SHGC value, first complete a CF1R-ENV-02-E to calculate the area-weighted average SHGC, which must be 0.23 or less, and attach it to the CF1R-ADD-01-E.

NOTES: (1) For the exceptions - up to 3 square feet (ft²) of tubular skylights and up to 16 square feet (ft²) of skylight area, enter 0.30.

(2) For the exception – up to 3 square feet (ft²) of glass in door, enter 0.23 for climate zones 2, 4, and 6-15 and enter 0.20 for climate zone 15; otherwise N/A.

12. Source: The source of the solar heat gain coefficient (SHGC) data for the fenestration product—indicate whether National Fenestration Rating Council (NFRC), Tables 110.6-A and 110.6-B, Equations NA6-1 and NA6-2, or Area-weighted Average Worksheet (CF1R-ENV-02).

13. Exterior Shading Device: If exterior shading devices are used to meet the solar heat gain coefficient (SHGC) requirement, indicate the type of device (from Table S-1 of CF1R-ENV-03-E Solar Heat Gain Coefficient Worksheet) and attach the CF1R-ENV-03-E.

NOTES: (1) An exterior shading device is not used for products with a National Fenestration Rating Council (NFRC) rated U-factor and SHGC based on a factory integrated shading device.

(2) Chromogenic glazing shall be considered separately from other fenestration.

14. Combined SHGC from CF1R-ENV-03: If exterior shading devices are combined with the solar heat gain coefficient (SHGC) value of the fenestration to meet the prescriptive SHGC requirements (as indicated in I. 13), indicate the SHGC calculated on form CF1R-ENV-03 and attach the form for each window with an exterior shading device.

15.-32. Automatically completed entries; no user input required.

J. Opaque Swinging Doors to Exterior

1. Tag/ID: Provide a name or designator for each unique door. This designator should be used consistently throughout the plan set (elevations, door schedules, etc.)
2. Area: Calculated area in square feet (ft²) for each unique door.
3. Proposed U-factor: Enter the proposed U-factor. If value is greater than 0.20, column 06 will autocomplete as Yes.
4. Source: National Fenestration Rating Council (NFRC) or Reference Appendices, Joint Appendix, Table 4.5.1s, 110.6-A and 110.6-B, Equations NA6-1 and NA6-2, or Area-Weighted Average Worksheet (CF1R-ENV-02).
5. Required Maximum U-factor. This field will always be 0.20.
6. Weighted Average: If column 03 is greater than 0.20 U-factor, attach form CF1R-ENV-02-E.
7. Comments: Any notes regarding location, unique conditions, or attachments.

K. Space Conditioning Systems – Heating/Cooling

If an existing space system will condition an addition, the prescriptive requirements do not apply to that system (Exception 4 to Section 150.2(a)). The enforcement agencies may require verification that the capacity of the existing heating system is adequate to meet the added load of the additional conditioned floor area. Since there is no health and safety code requirement to provide cooling, the enforcement agency will not ask for verification that the capacity of the existing system is adequate to meet the added load of the additional conditioned floor area.

If a new system is installed complete a Certificate of Compliance for Alterations to Space Conditioning Systems (CF1R-ALT-02).

1. Dwelling Unit Name: Name of dwelling unit or any other identifying name.
2. Dwelling Unit Total Conditioned Floor Area (CFA) – Sum of Existing Plus Addition (ft²): Total dwelling unit conditioned floor area in square feet (ft²), as measured from the outside of exterior walls of the dwelling unit or building being altered.
3. Comments: Any notes regarding location or unique conditions.

L. Water Heating Systems for Additions

Water heating compliance for an addition is described in Section 150.2(a)1D. When a water heater is added as part of an addition in a single dwelling the Prescriptive Standards allow ~~four~~three options under Section 150.2(a)1D.

1. Water Heating System Identification or Name: Provide a unique name for each unique water heating system type in the building. If the same water heating system type is used in more than one location in the building, it is sufficient to list the unique water heating system type only once. In order for one water heating system type to be considered the same as another, it must have the same description in Columns 2 through 12.
- ~~2. Water Heating System Type: Domestic Hot Water (DHW), Hydronic, or Combined Hydronic. DHW is for domestic hot water, hydronic is a water heating system used for space heating only; combined hydronic are when the water heater will provide both space conditioning and domestic hot water.~~
- ~~3-2.~~ 2. System option:
 1. A single heat pump water heater. The storage tank shall not be located outdoors and shall be placed on an incompressible, rigid insulated surface with a minimum thermal resistance of R-10. The water heater shall be installed with a communication interface that meets either the requirements of Section 110.12(a) or has an ANSI/CTA-2045-B communication port; or
 2. A single heat pump water heater that meets the requirements of NEEA Advanced Water Heater Specification Tier 3 or higher; or
 3. A gas or propane instantaneous water heater with an input of 200,000 Btu per hour or less and no storage tank; or
 4. For addition that are 500 square feet or less, an ~~instantaneous~~ electric water heater with point of use distribution as specified in the Reference Appendices, Residential Appendix, RA4.4.5
- ~~Water Heater Type: Electric water heater, Heat pump water heater, or NEEA Tier 3 or higher heat pump water heater.~~
- ~~4-3.~~ 3. Number of Water Heaters/Compressors in System:
- ~~5-4.~~ 4. Water heater Type: Field is auto-filled based on which system option was chosen.
- ~~6-5.~~ 5. Fuel Type: Field is auto-filled based on which system option was chosen. ~~For gas fired systems, please indicate whether the fuel source is natural gas or propane.~~
- ~~7-6.~~ 6. Tank Location: Field is auto-filled based on which system option was chosen.
- ~~8. Rated Input: Field is auto-filled based on which system option was chosen.~~
- ~~9-7.~~ 7. Distribution Type: Field is auto-filled based on which system option was chosen.

M. Indoor Air Quality (IAQ) Fan Information

1. Fan Name: Provide a unique name for each unique indoor air quality (IAQ) fan type in the building.
2. IAQ Type: Supply, ~~Exhaust~~, Balanced, Balanced – ERV, Balanced – HRV, Central Fan Integrated (CFI), Central Ventilation System – Supply, ~~Central Ventilation System – Exhaust~~, or Central Ventilation System – Balanced.
3. Comments: Any notes regarding location or unique conditions.

Documentation Declaration Statements

1. The person who prepared the CF1R will sign and complete the fields for their name, company (if applicable), address, phone number, certification information (if applicable), date and signature (may be electronic).
2. The person who is assuming responsibility for the project being built to comply with Title 24, Part 6, will complete the fields for their name, company (if applicable), address, phone number, license number (if applicable), date and signature (may be electronic).

Registration

The CF1R must be registered with a ~~HERSECC~~ provider prior to submitting for a building permit.

For information and data collection
only. Not valid until registered with an
ECC provider

A. General Information					
01	Project Name:	<<User Input: Text>>	02	Date Prepared:	<<User Input: Date>>
03	Project Location:	<<User Input: String>>	04	Building Front Orientation (deg):	<<User Input: IntegerNonnegative>>
05	CA City:	<<User Input: String>>	06	Number of Dwelling Units with Additions:	<<User Input: IntegerNonnegative; note if A11=Single Family, then the Fixed value=1>>
07	Zip Code:	<<User Input: Zipcode>>	08	Fuel Type:	<<User selects from list: Fixed value = Electricity, Natural Gas, Propane>>
09	Climate Zone:	<<User selects from list: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16>>	10	Total Conditioned Floor Area (ft²) (Addition):	<<User Input: IntegerNonnegative>>
11	Building Type	<< user select one from list: * Fixed value = Single Family >>	12	Slab Area (ft²):	<<User Input: IntegerNonnegative>>
13	Project Scope:	<< user select as many as are applicable from list: *Addition ≤ 300 ft² *Addition > 300 to ≤ 400 ft² *Addition > 400 to ≤ 700 ft² *Addition > 700 to ≤ 1000 ft² *ADU Addition ≤ 300 ft² *ADU Addition > 300 to ≤ 400 ft² *ADU Addition > 400 to ≤ 700 ft² *ADU Addition > 700 to ≤ 1000 ft² *Kitchen remodel *Space Heating System * Space Cooling System * Space Conditioning Duct System *Water Heating *Fenestration (if selected, show A14) *Opaque Exterior Doors>>	14	Fenestration Exceptions:	<<User selects as many as are applicable from list: *NA (do not allow other entries) *Installing ≤ 3ft² glass in door *Installing ≤ 3ft² tubular skylight *Installing ≤ 16ft² skylight

B. Opaque Surface Details – Framed Building Insulation Details – Framed Walls/ Framed Floors (Section 150.2(a) and 150.1(c)1)

<<if A13 = one of the items in the following list:

*Addition ≤ 300 ft2

*Addition > 300 to ≤ 400 ft2

*Addition > 400 to ≤ 700 ft2

*ADU Addition ≤ 300 ft²

*ADU Addition > 300 to ≤ 400 ft²

*ADU Addition > 400 to ≤ 700 ft²

*ADU Addition > 700 to ≤ 1000 ft²

*Addition > 700 to ≤ 1,000 ft2 then display this section ~~but entry is optional~~,

else display the standard “This Section Does Not Apply” message>>

01	02	03	04	05	06a	06b	07	08	09	1008	1109
Tag/ID	Assembly Type	Frame Type	Frame Depth (inches)	Frame Spacing (inches)	Proposed			Appendix JA4 Reference		Required	Comments
					Cavity R-value	Continuous Insulation R-value	U-Factor	Table	Cell	U-Factor from Table 150.1-A	
<<User Input: Object Name Permissible>>	<<User selects from list: Ceiling, Floor, Roof, Wall or Attic>>	<< User selects from Wood or Metal>>	<< User selects from 2x4, 2x6, 2x8, 2x10, 2x12 or 2x14; else value = NA >>	<< user selects from list: 16InchOC, or 24InchOC>> else value = NA >>	<<User Input: Decimal Nonnegative>>	<<User Input: Decimal Nonnegative>>	<<User Input: Decimal Nonnegative>>	<<User Input: JA_TableID >>	<<User Input: JA_TableCell>>	<<User Input: Decimal Nonnegative>>	<<User Input: Text>>

Note:
<<Static Text - see the top>>

C. ~~Opaque Surface~~Building Insulation Details – Nonframed (Section 150.1(c)1)

<<if A13 = one of the items in the following list:

*Addition ≤ 300 ft²

*Addition > 300 to ≤ 400 ft²

*Addition > 400 to ≤ 700 ft²

*ADU Addition ≤ 300 ft²

*ADU Addition > 300 to ≤ 400 ft²

*ADU Addition > 400 to ≤ 700 ft²

*ADU Addition > 700 to ≤ 1000 ft²

*Addition > 700 to ≤ 1,000 ft² then display this section ~~but entry is optional,~~
else display the standard “This Section Does Not Apply” message>>

01	02	03	04	05	06	07	08	09	1008	1109
Tag/ID	Assembly Type	Assembly Materials	Thickness (inches)	Proposed			Joint Appendix JA4 Reference		Required	Comments
				Core Insulation R-value	Continuous Insulation R-value	U-Factor	Table	Cell	U-Factor from Table 150.1-A	
				Core Insulation R-value	Continuous Insulation R-value	U-Factor	Table	Cell	U-Factor from Table 150.1-A	
<<User Input: Object Name Permis sive>>	<<User selects from list: Roof, Wall>>	<<User selects from list: *SIP OSB, *SIP 2x, *SIP 4x, *SIP I-Joist, *Metal Panel Walls, *Log Home Walls, *Straw Bale Walls, *Insulating Concrete Form >>	<<User Input: DecimalN onnegative e>>	<<User Input: DecimalNo nnegative> >	<<User Input: DecimalNo nnegative> >	<<User Input: DecimalN onnegativ e>>	<<User Input: JA_Tabl eID>>	<<User Input: JA_Tabl eCell>>	<<User Input: DecimalNonne gative>>	<<User Input: Text>>

Note:

<<Static Text - see the top>>

D. Opaque Surface Building Insulation Details – Masonry/Mass Walls (Section 150.1(c)1Bii)

<<if A13 = one of the items in the following list:

*Addition ≤ 300 ft2

*Addition > 300 to ≤ 400 ft2

*Addition > 400 to ≤ 700 ft2

*ADU Addition ≤ 300 ft2

*ADU Addition > 300 to ≤ 400 ft2

*ADU Addition > 400 to ≤ 700 ft2

*ADU Addition > 700 to ≤ 1000 ft2

*Addition > 700 to ≤ 1,000 ft2 then display this section ~~but entry is optional~~,
else display the standard “This Section Does Not Apply” message>>

01	02	03	04	05	06	0705	0806	0907	1008	11	12	1309	1410	1511	1612
Tag/ID	Above or Below Grade?	Masonry/ Mass Type	Masonry/ Mass Thickness (inches)	Proposed						Required					
				Mass-Wall Joint Appendix JA4 Reference		Exterior Insulation		Interior Insulation		Insulation Layer Joint Appendix JA4 Reference		Exterior Insulation		Interior Insulation	
				Table	Cell	R-value	U-factor	R-value	U-factor	Table	Cell	R-value	U-factor	R-value	U-factor
<<User input: ObjectNamePermissive>>	<<User select from list: *Above Grade; or *Below Grade>>	<<User selects from list: *Clay Brick, *Clay Hollow Unit, *CMU Light Weight, *CMU Medium Weight, *CMU Normal Weight, *Concrete	<<User input: Decimal Nonnegative>>	<<User input: JA_TableID>>	<<User input: JA_TableCell>>	<<User input: Decimal Nonnegative; Note: Either R-value or U-factor is required, not both. The same descriptor should be used throughout this table>>	<<User input: Decimal Nonnegative; Note: Either R-value or U-factor is required, not both. The same descriptor should be used throughout this table>>	<<User input: Decimal Nonnegative; Note: Either R-value or U-factor is required, not both. The same descriptor should be used throughout this table>>	<<User input: Decimal Nonnegative; Note: Either R-value or U-factor is required, not both. The same descriptor should be used throughout this table>>	<<User input: JA_TableID>>	<<User input: JA_TableCell>>	<<User input: Decimal Nonnegative; Note: Either R-value or U-factor is required, not both. The same descriptor should be used throughout this table>>	<<User input: Decimal Nonnegative; Note: Either R-value or U-factor is required, not both. The same descriptor should be used throughout this table>>	<<User input: Decimal Nonnegative; Note: Either R-value or U-factor is required, not both. The same descriptor should be used throughout this table>>	<<User input: Decimal Nonnegative; Note: Either R-value or U-factor is required, not both. The same descriptor should be used throughout this table>>

Note:
<<Static Text - see the top>>

E. Slab On Grade/Concrete Raised Floor Insulation (Table 150.1-A)

<<if A12 > 0 AND A13 = one of the items in the following list:

*Addition ≤ 300 ft2

*Addition > 300 to ≤ 400 ft2

*Addition > 400 to ≤ 700 ft2

*ADU Addition ≤ 300 ft²

*ADU Addition > 300 to ≤ 400 ft²

*ADU Addition > 400 to ≤ 700 ft²

*ADU Addition > 700 to ≤ 1000 ft²

*Addition > 700 to ≤ 1,000 ft2 then display this section;

else display the standard “This Section Does Not Apply” message>>

01	02	03	04	05	06
Floor Type	Proposed		Required		Comments
	Insulation R-value	Insulation U-factor	Insulation R-value	Insulation U-factor	
<<User select from list: *R-0; *R-4; *R-5; *R-7; *R-8; *R-10 vertical; *R-10 vertical and R-7 horizontal; or *NA; If E01 = ‘Heated slab’ or A09 = 16, then ‘NA’ not allowed Note: Either R-value or U-factor is required, not both>> <<User selects from list: Slab on grade, Concrete raised, Heated Slab >>	<<User select from list: *R-0; *R-4; *R-5; *R-7; *R-8; *R-10 vertical; *R-10 vertical and R-7 horizontal; or *NA; If E01 = ‘Heated slab’ or A09 = 16, then ‘NA’ not allowed Note: Either R-value or U-factor is required, not both>>	<<If E01 = ‘Heated slab’ or A09 = 16, then require User Input: DecimalNonnegative; else allow NA Note: Either R-value or U-factor is required, not both>>	<<if E02 = NA, then value = NA; elseif A09 = 16 and E01 = “Slab on grade”, then value = R-7; Else if A09 = 1-15 and E01 = “Slab on grade”, then value = NA; elseif A09 = 1-2, 11, 13-14, or 16 and E01 = “Concrete raised”, then value = R-8; elseif A09 = 3-10 and E01 = “Concrete raised”, then value = R-0; elseif A09 = 12 or 15 and E01 = “Concrete	<<if E03 = NA, then value = NA; elseif A09 = 16 and E01 = “Slab on grade”, then value = 0.58; Else if A09 = 1-15 and E01 = “Slab on grade”, then value = NA; elseif A09 = 1-2, 11, 13-14, or 16 and E01 = “Concrete Raised”, then value = 0.092; elseif A09 = 3-10 and E01 = “Concrete Raised”, then value = 0.269; elseif A09 = 12 or 15 and E01 = “Concrete	<<User Input: Text>>

			raised”, then value = R-4; elseif A09 = 1-15 and E01 = “Heated slab”, then value = R- 5; elseif A09 = 16 and E01 = “Heated slab”, user selects from *R-10 vertical or *R-10 vertical and R-7 horizontal>>	Raised”, then value = 0.138; elseif E01 = “Heated slab”, then value = NA Note: Range check, value in E03 must be less than or equal to E05 (both can be NA) in order to comply>>	
Note: <<Static Text - see the top>>					

F. Radiant Barrier (Section 150.1(c)2)

<<A09 = 2 – 15 AND if A13 = one of the items in the following list:

*Addition ≤ 300 ft2

*Addition > 300 to ≤ 400 ft2

*Addition > 400 to ≤ 700 ft2

*ADU Addition ≤ 300 ft2

*ADU Addition > 300 to ≤ 400 ft2

*ADU Addition > 400 to ≤ 700 ft2

*ADU Addition > 700 to ≤ 1000 ft2

*Addition > 700 to ≤ 1,000 ft2 then display this section ~~but entry is optional~~,
else display standard “This Section Does Not Apply” message>>

01	02
Radiant Barrier installed below the roof deck and on all gable end walls	Comment
<< user selects from list: Yes, No, NA >> <<if No is selected, display ‘Do NOT PROCEED’>>	<<if F01 = NA, require user Input: Text; Else optional input>>
<<Static Text - see the top>>	

G. Roofing Products (Cool Roof) (Section 150.1(c)11)

<<If A13= one of the items in the following list:

*Addition > 300 to ≤ 400 ft2

*Addition > 400 to ≤ 700 ft2

*ADU Addition > 300 to ≤ 400 ft2

*ADU Addition > 400 to ≤ 700 ft2

*ADU Addition > 700 to ≤ 1000 ft2

*Addition > 700 to ≤ 1000 ft2 then display this section ~~but entry is optional~~,
else display the standard “The Section Does Not Apply” message>>

01	02	03	04	05	06	07	08	09	10	11	12	13
Tag/ID	Exception	Roof Pitch	Method of compliance	Product Type	CRRC Product ID Number	Proposed				Required		
						Initial Solar Reflectance	Aged Solar Reflectance	Thermal Emittance	SRI (Optional)	Aged Solar Reflectance	Thermal Emittance	SRI (Optional)
<<User Input: Object Name Permissive>>	<<User selects from list: 1, 2, or None; If 1 or 2, value = “Meets	<<User selects from list: Roof pitch is ≥ 2:12 or Roof	<< User selects from list: Not in an applicable climate zone, Aged	<<User selects from CRRC product list: Asphalt Shingles,	<<User input From the CRRC Directory if user knows what they are going to install. Else allow user to enter NA>>	<<if G04 = ‘Not in an applicable climate zone’ then result = NA;	<<if G04 = ‘Not in an applicable climate zone’ then result = NA; else User Input:	<<if G04 = ‘Not in an applicable climate zone’ then result = NA; else User	<<if G04 ≠ ‘SRI’ then result is NA; Else user enters value from	<< If A09 = 13 or 15 and G03 = < 2:12, then value = 0.63; elseif A09 = 10-15 and	<<If A09 = 13 or 15 and G03 = < 2:12 and G04 = ‘SRI’, then value = 0.85; if G04 is ‘Aged Solar	<< if A09 = 13 or 15 and G03 = < 2:12 and G04 = ‘SRI’, then

	cool roof requirem ents"; Elseif None, go to G03>>	pitch is < 2:12>>	Solar Reflectanc e and Thermal Emittance, or SRI>>	Built-up Roofing, Clay Roof Tiles, Factory Applied Coating, Field Applied Coating, Metal Roof, Modified Bitumin Sheet, Roof Pavers, Single Ply Thermopla stic, Single Ply Thermoset, Metal Shakes Shingles Fluid Applied Membrane Polymer Composite Steep Slope, Spray Polyuretha ne Foam, Stone Aggregate Ballast>>		Else If the user knows what they are installing, the user enters the DecimalNonn egative (x.xx, value must be > 0 and < 1)>>	DecimalNonn egative (x.xx, value must be > 0 and < 1)>>	Input: DecimalNon negative (x.xx, valu must be > 0 and < 1)>>	CRRC directory or from a completed SRI Worksheet >>	G03 = ≥ 2:12 then value = 0.20; else value = NA >>	Reflectance and Thermal Emittance' then value = 0.75; if A09 = 10-15 and G03 = ≥ 2:12, then value = 0.75; Else for all other combinations, value = NA>>	value = 75; if A09 = 10 – 15 and G03 = ≥ 2:12, then value = 16; Else for all other combinati ons, value = NA>>

Note:
<<Static Text - see the top>>

H. Fenestration/Glazing Allowed Areas and Efficiencies (Section 150.2(a)1)

<<if A13 contains Fenestration then display this section; else display the standard “This Section Does Not Apply” message>>

01	02	03	04	05	06	07	08	09	10
Addition Type ft²	Maximum Allowed Fenestration Area For All Orientations ft²		Maximum Allowed West-Facing Fenestration Area Only ft²		Maximum Allowed U-factor (Windows)	Maximum Allowed U-factor (Skylights)	Maximum Allowed SHGC (Windows)	Maximum Allowed SHGC (Skylights)	Comments
	The Greater		The Greater						
	Maximum Calculated based on Allowed %	Maximum Calculate d Allowed ft²	Maximum Calculated based on Allowed %	Maximum Calculated Allowed ft²					
<This value is auto-filled with the Project Scope in section A: *Addition ≤ 300 ft² *Addition > 300 to ≤ 400 ft² *Addition > 400 to ≤ 700 ft² *Addition > 700 to ≤ 1,000 ft² *ADU Addition ≤ 300 ft² *ADU Addition > 300 to ≤ 400 ft² *ADU Addition > 400 to ≤ 700 ft² *ADU Addition > 700 to ≤ 1000 ft²>>	<<Calculated Value: If H01 equals Addition ≤ 300 ft², Addition > 300 to ≤ 400 ft², ADU Addition ≤ 300 ft², or ADU Addition > 300 to ≤ 400 ft², value equals A10 * 0.30; Elseif Addition > 400 to ≤ 700 ft² or ADU Addition > 400 to ≤ 700 ft², value equals A10 * 0.25; Elseif Addition > 700 to ≤ 1000 ft² or ADU Addition > 700 to ≤ 1000 ft², value equals A10 * 0.20>>	<<Calculated Value: If H01 equals Addition ≤ 300 ft² or Addition > 300 to ≤ 400 ft², value equals 75ft²; Elseif Addition > 400 to ≤ 700 ft², value equals 120ft²; Elseif Addition > 700 to ≤ 1000 ft², value equals 175ft²>>	<<Calculate d Value: if A09 = 1, 3, 5 or 16, then value equals NA; Else if H01 equals Addition ≤ 300 ft², Addition > 300 to ≤ 400 ft², or Addition > 400 to ≤ 700 ft², then value equals NA; Else if Addition > 700 to ≤ 1000 ft², value equals (A10*0.05)> >	<<Calculate d Value: If A09 = 1, 3, 5 or 16, then value equals NA; Else if H01 equals Addition ≤ 300 ft², Addition > 300 to ≤ 400 ft² or Addition > 400 to ≤ 700 ft², then value equals 60ft²; Elseif Addition > 700 to ≤ 1000 ft², value equals 70ft²>>	<< If A09 equals 1 through 5, 11 through 14 and 16 then report value = 0.30>0.27; Elseif A09 equals 6 through 10 and 15 then report value = 0.30 >>	<<If A14 contains Installing ≤ 3ft² tubular skylight, or Installing ≤ 16ft² skylight, if A09 equals 2, 4, and 6 through 15 then report 0.5540; Elseif A09 equals 1 through 5, 11 through 14 and 16 then report 0.27; equals 1, 3, 5, 16 then report 0.27N/A; Else report value = 0.30>>	<<If A09 equals 1, 3, 5 or 16 then report N/A; If A09 = 15 then value = 0.20; Else value = 0.23; >>	<< If A09 = 1, 3, 5 or 16, then result equals NA; Else if A14 contains Installing ≤ 3ft² tubular skylight, or Installing ≤ 16ft² skylight, then report 0.30; Elseif A09 equals 1, 3, 5, 16 then report NA; Else report value = 0.23>>	<<User Input: Text>>

I. Fenestration Proposed Areas and Efficiencies

<<if A13 contains Fenestration then display this section; else display the standard “This Section Does Not Apply” message>>

01	02	03	04	05	06	07	08	09	10	11	12	13	14
Tag/ ID	Fenestration Type	Frame Type	Dynamic Glazing	Orientation N, S, W, E	Number of Panes	Proposed Fenestration Area (ft²)	Proposed West Facing Fenestration Area (ft²)	Proposed U-factor	Proposed U-factor Source	Proposed SHGC	Proposed SHGC Source	Exterior Shading Device	Combined SHGC from CF1R-ENV-03
<< User Input: String (max 50 characters)>>	<< User selects from list: *fixed window, *operable window, *skylight *tubular skylight (only allow if A14 contains Installing ≤ 3 ft² tubular skylight), *glass in door (only allow if A14 contains Installing ≤ 3 ft² glass in door)>>	<< User selects from list: Metal, Metal Thermal Break and non-metal>>	<< User selects from list: None, Chromogenic Glazing, Integrated Shading device>>	<<User selects from list: North, East, South and West>>	<<User selects from list: Single pane, Double pane and Triple pane>>	<< If I05 equals North, East or South, then enter user defined number in this column>>	<< If I05 equals West, then enter user defined number in this column>>	<<User Input: DecimalNonnegative>>	<<User selects from list: NFRC, Tables 110.6-A and 110.6-B, Equations NA6-1 and NA6-2, or Area-weighted Average Worksheet (ENV-02)>>	<<User Input: DecimalNonnegative; <u>Else</u> <u>If H08 or H09 equals NA then report NA</u> >>	<< <u>If I11 equals NA then report NA</u> ; <u>Else</u> <u>If user selects from list: NFRC, Tables 110.6-A and 110.6-B, Equations NA6-1 and NA6-2, or Area-weighted Average Worksheet (ENV-02)>></u>	<<User selects from list: Drop Arm Awnings, Louvered Sun Screen, Low Sun Angle Louvered Sun Screen, None, Operable Awnings, South-facing Overhang, Retractable Awnings, Roll Down Blinds Or Slats, Standard Bug Screen, Sun Screen, Vertical Roller Or Shades>>	<<If I13 equal to None, or I04 contains Chromogenic glazing, then report N/A; Else report value from CF1R-ENV-03, <u>field D04</u> ; <u>Else If H08 or H09 equals NA then report NA</u> >>

CERTIFICATE OF COMPLIANCE – DATA FIELD DEFINITIONS AND CALCULATIONS	CF1R-ALT-01-E
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15	Total Proposed Fenestration Area	<<calculate (sum of column I07 and I08)>>
16	Maximum Allowed Fenestration Area	<<Report Greatest Value from H02 or H03>>
17	Compliance Statement	<<If I15 is equal to or less than I16 then report: Design Complies with the Total Allowed Fenestration Area Else report: Total Proposed Fenestration Area Exceeds Allowable, Do Not Proceed >>
18	Total Proposed West-Facing Fenestration Area	<<calculate (sum of column I08)>>
19	Maximum Allowed West-Facing Fenestration Area	<< Report Greatest Value from H04 or H05>>
20	Compliance Statement	<<If I18 is equal to or less than I19, or I19 equals N/A, then report: Design Complies with the Total Allowed West-facing Fenestration Area; Else report: Total Proposed West-facing Fenestration Area Exceeds Allowable, Do Not Proceed >>
21	Proposed Fenestration U-factor (Windows)	<<If I02 contains fixed or operable window and if all associated values listed in column I09 are less than or equal to H06 then enter the largest value from list; If I02 does not contain fixed or operable window then value equals NA; Else enter the weighted average value from the CF1R-ENV-02>>
22	Required Fenestration U-factor (Windows)	<< Report Value from H06>>
23	Compliance Statement	<<If I21 equals NA then value equals NA; If I21 is equal to or less than I22 then report: Design Complies with the Maximum Allowed Fenestration U-value; Else report: Fenestration U-value Exceeds Allowable, Do Not Proceed >>
24	Proposed Fenestration SHGC (Windows)	<<If I02 contains fixed or operable window and if all associated values listed in column I11 or I14 are less than or equal to 0.23 then report the single largest value from the two lists;

			<u>Else If I11 or I14 equals NA, then report NA;</u> If I02 does not contain fixed or operable window then value equals NA; Else enter the weighted average value from the CF1R-ENV-02>>
25	Required Fenestration SHGC (Windows)		<<Report Value from H08>>
26	Compliance Statement	<<If I24 equals NA then value equals NA; If I24 is equal to or less than I25, or I25 equals N/A, then report: Design Complies with the Maximum Allowed Fenestration SHGC; Else report: Fenestration SHGC Exceeds Allowable, Do Not Proceed>>	
27	Proposed Fenestration U-factor (Skylights)		<<If I02 contains skylights then enter the largest value from I09; If I02 does not contain skylights then value equals NA>>
28	Required Fenestration U-factor (Skylights)		<< Report Value from H07>>
29	Compliance Statement	<<If I27 equals NA then value equals NA; If I27 is equal to or less than I28 then report: Design Complies with the Maximum Allowed Fenestration U-value; Else report: Fenestration U-value Exceeds Allowable, Do Not Proceed>>	
30	Proposed Fenestration SHGC (Skylights)		<<If I02 contains skylight then enter the single largest associated value from columns I11 or I14; <u>Else If I11 or I14 equals NA, then report NA;</u> If I02 does not contain skylight then value equals NA>>
31	Required Fenestration SHGC (Skylights)		<<Report Value from H09; <u>Else If I11 or I14 equals NA, then report NA;</u>>>
32	Compliance Statement	<<If I30 equals NA then value equals NA; If I30 is equal to or less than I31 then report <u>or I31 equals NA</u>: Design Complies with the Maximum Allowed Fenestration SHGC;	

		Else report: Fenestration SHGC Exceeds Allowable, Do Not Proceed>>
Note: <<Static Text - see the top>>		

For information and data collection only. Not valid until registered with an ECC provider

J. Opaque Swinging Doors to Exterior (Section 150.1(c)5) <<if A13 contains Opaque Exterior Doors, then display this section; else display the standard “This Section Does Not Apply” message>>						
01	02	03	04	05	06	07
Tag/ID	Area	Proposed U-factor	Proposed U-factor Source	Required Maximum U-factor	Weighted Average (Yes/No)	Comments
<<user input: string (max 50 characters)>>	<<user input: DecimalNonnegative xxx.xx>>	<<user input: DecimalNonnegative x.xxx>>	<<user pick from list: *NFRC; or *Area-weighted Average Worksheet (ENV-02)>>	<<default value = 0.20>>	<<if J03 > 0.20, then value = Yes; Else value = No>>	<<user input: text>>
Note: <<Static Text - see the top>>						

K. Space Conditioning (SC) Systems – Heating/Cooling (Section 150.2(b) or (Section 150.1(c)7) << if none of the following are selected in A13: Space Heating System, or Space Cooling System, or Space Conditioning Duct System, then display the section does not apply message; else if one or more of the following are selected in A13: Space Heating System, or Space Cooling System, or Space Conditioning Duct System, then require one row of data in this table>> <<the data registry shall require a CF1R-ALT-02 for the dwelling>>		
01	02	03
Dwelling Unit Name	Dwelling Unit Total CFA = Sum of Existing + Addition (ft ²)	Comments
<<User Input: ObjectNamePermissive>>	<<User Input: IntegerNonnegative>>	<<User Input: Text>>

L. Water Heating Systems (Section 150.2(a)1D)

<<This section is required if A13 ProjectScopeAddition includes Water Heating Addition; else display the standard “This Section Does Not Apply” message >>

01	02	0302	0403	0504	0605	07	0806	0907
Water Heating System ID or Name	Water Heating System Type	System Option (from §150.2(a)1D)	# of Water Heaters/ Compressors in System	Water Heater Type	Fuel Type	Rated Input	Tank Location	Distribution Type
<<user input text>>	<<user pick from list: *DHW; *Hydronic; *Combined Hydronic>>	<<user pick from list: 1, 2, 3, 4>>	<<user input: nonnegative number>>	<<if L03 =1 then value = Heat pump water heater; if L03 = 2 then value = NEEA Tier 3 heat pump water heater; if L03 = 3 or 4, then value = Consumer Instantaneous electric water heater>>	<<if L03 = 1 or 2 then value = Heat pump; if L03 = 3 4 then value = Electricity; if L03 = 3, then user pick from list: *Natural Gas, *Propane>>	<<if L03 = 3 then value = ≤ 200,000; else value = N/A>>	<<if L03 = 1 or 2, user select from list: *Garage, or *Conditioned space; else value = N/A>>	<<if L03 = 4 then value = Point of Use Distribution; else value = Standard>>

Options:

<<Static Text - see the top>>

M. Indoor Air Quality (IAQ) Fan Information

<<If A13= one of the items in the following list:

*ADU Addition ≤ 300 ft²

*ADU Addition > 300 to ≤ 400 ft²

*ADU Addition > 400 to ≤ 700 ft²

*ADU Addition > 700 to ≤ 1000 ft² then display this section,

else display the standard “The Section Does Not Apply” message>>

01	02	03
Fan Name	IAQ Type	Comments
<<user input text>>	<<User pick from list: **Supply **Exhaust; or **Balanced; or **Balanced – ERV; or **Balanced – HRV; or **Central Fan Integrated (CFI); or **Central Ventilation System – Supply; or **Central Ventilation System – Exhaust; or **Central Ventilation System – Balanced>>	<<user input text>>